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Cover image: The ambiguous image, “My Wife and My Mother-in-Law”, by the cartoonist W. E. Hill, 1915 (adapted from a picture going back at least to an 1888 German postcard). The image shows a young woman with her face turned away, or the profile of an old woman. It illustrates that people are capable of perceiving multiple iconicities at once.

Bonnie McLean

Arbitrary by Accident, Iconic by Design

On the role of motivation in the nature and development of lexicons

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Abstract
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This thesis investigates the role of iconicity, or motivated relations between meaning and form, in the nature and development of lexicons. Papers I and II develop a perspective on iconicity as a subjective process of construal, and detail how a robustness approach can meet the challenges this poses for the operationalisation of iconicity in scientific investigations. Paper III shows how iconicity can act as a factor of attraction affecting stability and change in the development of form-meaning mappings in lexicons, with an experiment exploring the transmission of artificial languages in an iterated learning and interaction paradigm. We see how generations of participants reshape arbitrary languages to become more iconic, and how iconicity is maintained in iconic languages over several generations. The results are explained by iconic mappings being easier to learn and to co-ordinate on. However, it should not be taken from this that iconicity is simple. Papers IV and V show iconicity in natural language is much more complex than it appears in experiments, with an in-depth analysis of ideophones. Paper IV is an fNIRS study of Japanese ideophones, which reveals them to be perfectly honed tools for a variety of complex linguistic, perceptual, and sociopragmatic functions. Paper V explores the challenges ideophones pose for documentary linguistics, presenting a mixed methods toolkit for eliciting, documenting, and analysing ideophones in a way that makes sense of their sophisticated semiotics. I show how, taken together, the papers in the thesis encourage us to reconsider traditional perspectives on language structure that see arbitrariness as the dominant principle, and iconicity as auxiliary. Such perspectives are grounded in a concept of iconicity that is misinformed, and incompatible with the account of iconicity developed in this thesis. With an evolutionary approach, the centrality of iconicity to language structure, and the marginal contributions of arbitrariness become even starker. I argue that it is iconicity that is built into the design of language, while arbitrariness may be merely an accident.

Keywords: iconicity, language evolution, language change, ideophones

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To Toni Mummery

List of Papers

This thesis is based on the following papers, which are referred to in the text by their Roman numerals.

I. McLean, Bonnie and Yasamin Motamedi. 2026. "A robustness approach to operationalisations of iconicity." In Fischer, O., Akita, K., & Perniss, P. (eds.), *The Oxford Handbook of Iconicity in Language* (pp. 659–677). Oxford University Press.

II. McLean, Bonnie, Dunn, Michael and Mark Dingemanse. 2023 "Two measures are better than one: combining iconicity ratings and guessing experiments for a more nuanced picture of iconicity in the lexicon." *Language and Cognition*, *15*(4), 716–739.

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IV. McLean, Bonnie, Akita, Kimi and Dan Dewey. "Not just for play: fNIRS reveals complex linguistic, perceptual, and sociopragmatic functions of Japanese ideophones." *Under Review.*

V. McLean, Bonnie and Mark Dingemanse. 2025. "A multi-methods toolkit for documentary research on ideophones." In Jeffrey P. Williams (ed.), *Capturing Expressivity: Contexts, Methods, and Techniques for Linguistic Research* (pp. 74–107). Oxford University Press.

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Glossary

Iconicity	A perceived resemblance between aspects of form and meaning (Dingemanse, 2019).
Indexicality	A causal or physical connection between form and meaning.
Systematicity	A statistical relation between patterns of forms and meanings.
Sound Symbolism	A cover term for motivated form-meaning relations in lexicons, subsuming iconicity, indexicality, and systematicity.
(Linguistic) Arbitrariness	(Of lexemes) a seemingly unmotivated (from a synchronic perspective) relation between form and meaning.
Ideophones	An open lexical class of marked words that depict sensory imagery (Dingemanse, 2012). As lexicalised depictions, ideophones convey meaning through their phonological forms, making use of features like iconicity, indexicality, and systematicity.
Depiction	An analogical mode of communication that invites and affords the construal of iconic mappings between form and meaning (Dingemanse, 2019).

1. Introduction

"I have often talked over this matter, both with Cratylus and others, and cannot convince myself that there is any principle of correctness in names other than convention and agreement."

Plato, *Cratylus,* 384d

The study of language has come a long way since Plato's *Cratylus*, yet when it comes to the debate around the nature of the connection between word meanings and forms, the arguments for Hermogene's position, now known as 'linguistic arbitrariness' (Saussure, 1916), have not evolved much beyond the statement exemplified in the above quotation that it is hard to see how it could be any other way. On the other hand, the arguments for Cratylus's position, that word-meaning relations can be motivated, have evolved considerably. We now have a much better understanding of what it means for language to be motivated, which is quite different from the deterministic, limiting picture of linguistic motivation reflected in Plato's dialogue. These advancements have implications for how we think about language structure, particularly when it comes to the role of arbitrariness—often framed as an answer to the inherent limitations of a fully motivated language.

At the same time as our knowledge of motivation in language has been evolving, so too has our knowledge of how language itself evolves, and here again we must question the relevance of the principle of arbitrariness. Saussure himself considered the arbitrariness of the linguistic sign to be trivial were it not for its effects on how language changes. He argued that the arbitrariness of the sign enables its stable transmission between generations of speakers, as succeeding generations have no "reasonable basis" on which to modify the signs they inherit (Saussure, 1916, p. 75). At the same time, he also saw it as responsible for the way in which languages change, appearing to be borne along "willy-nilly" by the currents of time (Saussure, 1916, p. 76). He contrasts this with the way in which other social institutions change by adapting "the mean employed to the ends", noting that arbitrariness precludes this kind

of adaptive change which is only possible in systems based "on the natural relations of things" (Saussure, 1916, p. 75). In short, Saussure argued that language evolves the way that it does because it is the way that it is. Under this view, linguistic arbitrariness is an explanatory principle.

However, another way of looking at it is that language is the way that it is because it evolves the way that it does, a perspective that has gained considerable traction in recent decades (Kirby et al., 2014; Tamariz & Kirby, 2016; Beckner et al., 2009). Under this view, the source of stability and change in language is found not within the system itself but within its users (Christiansen & Chater, 2008), for whom connecting meaning and form could not come more naturally. The system develops not only through drift but through biased transformation and selection in the process of language learning and use. This perspective leaves much less room for arbitrariness as an exploratory principle, as opposed to just something to be explained. It does, however, open up the possibilities for motivation to act as an explanatory factor in language change, and indeed early critics of arbitrariness argued their case from just this position (Bolinger, 1949; Jespersen, 1922a).

Despite these developments, the place of arbitrariness within linguistic theorising remains largely unquestioned, even as space is made for iconicity and other forms of motivation (e.g. Pleyer et al., 2025; Murgiano et al., 2021; Dingemanse et al., 2015). The willingness to reconsider the role of iconicity in language has not been matched by a willingness to reconsider the role of arbitrariness. Despite its provocative title, this thesis is not an answer to this problem. Papers I – V largely consider iconicity on its own terms. However, in developing the thesis it became clear that, as long as iconicity is framed as a counterpart to arbitrariness, it is all too easy for those uninterested in the subject to relegate it to the margins of linguistic enquiry, when it really belongs in the centre. For this reason, after summarising the findings from Papers I – V, I focus on their relevance to the argument that current perspectives on language tend to underestimate the contributions of iconicity and other forms of motivation to linguistic structure, while overestimating the contributions of arbitrariness (Jespersen, 1917; Waugh, 1994; Nuckolls, 1999).

Section 2 introduces the topic of motivation in lexicons, focusing the discussion around debunking three prominent myths—that motivation in language is deterministic, exceptional, and inherently limited in scope—that have allowed the idea of arbitrariness as a necessary complement to motivation in language to persist. These perspectives are defended with a variety of evidence, from behavioural data obtained in psycholinguistic experiments (Papers I, II), to neuroimaging data (Paper IV), and qualitative analysis of ideophones in natural languages (Paper V).

While arbitrariness often seems to dominate motivation in language from a synchronic perspective, when we take an evolutionary approach, this relation is reversed. This is highlighted in Section 3, which considers the role of motivation in language change. While the role for motivation in the formation of words is generally accepted, its contributions to their continued development are not so well explored. It is often assumed that, regardless of how a word began, once it enters into the linguistic system it functions only as a symbol, and it is the conventional mode of meaning making that dominates from there on out. I argue instead that iconic and indexical factors remain relevant, and shape how conventions shift, and that this is true for all words regardless of their origins. Given the right confluence of factors (a particular sound in close enough proximity to a particular meaning), motivated form-meaning associations can act as attractors (Sperber, 1985, 1996) that shape the development of form-meaning mappings in specific directions. Paper III provides an experimental demonstration of this.

Section 4 concludes the discussion by reiterating the central argument of the thesis that the attention given to arbitrariness in theories of language structure is unnecessary once the very real contributions of motivation are properly considered.

Summary of papers

Paper I introduces the topic of motivation in lexicons, and tackles the problem of how best to operationalise concepts like iconicity, systematicity, and indexicality in scientific analyses, to better understand their unique contributions to fitness between linguistic form and function. The conclusion is that while no measure is without bias, triangulating insights from multiple measures affords greater precision than any single measure alone (see also Punselie et al., 2024).

Paper II compares two common behavioural measures of iconicity—iconicity ratings and guessing accuracies—to better understand their relative strengths and weaknesses. We collected guessing accuracies and iconicity ratings for the same Japanese words from naive, English-speaking participants. Encouragingly, the two measures were correlated, suggesting that they do indeed capture the same phenomenon, i.e., iconicity (see also Winter & Perlman, 2021a). However, there were also differences between the measures which revealed iconicity was not the *only* thing each measure captured. Iconicity ratings were inflated for words that were structurally marked, highlighting the role that framing can play in iconic effects. Meanwhile guessing accuracies showed some words were guessed at rates significantly *below* rather than at or above chance, suggesting a role for language-specific experience in the

appreciation of the iconicity of some forms (Akita & Imai, 2022; Imai & Akita, 2023). The study highlights how operationalisations of iconicity, especially when used in combination, can provide insight into the nature of iconicity itself. We also introduce some modifications to the design of the guessing task that lead to improvements in the robustness, sensitivity, and discriminability of this measure, together with a reproducible workflow (*Icotools*) for generating rating and guessing tasks from standardised wordlists (McLean, 2022).

Having achieved a better grasp of iconicity, we turn our attention in Paper III to its counterpart, arbitrariness. We critically examine the notion of arbitrariness as a design feature of language, with an iterated communication game that introduces a framework for studying semantic change in a laboratory setting. A typical pattern in this type of study is for originally motivated forms to gradually lose their iconic qualities over repeated rounds of interaction (see Fay et al., 2014 for a review). However, such effects emerge from a pressure for compressibility, rather than a drive towards arbitrariness. Here we show that if you remove the compressibility pressure from the equation by keeping forms fixed, but allow participants to alter form-meaning mappings, what follows is a drive towards iconic and indexically motivated mappings, rather than a descent into arbitrariness. The reasons behind this are twofold: (1) motivated mappings are more learnable, and (2) motivated mappings are easier to coordinate on in interaction. We analyse these findings using the framework of cultural attractor theory (Sperber, 1985, 1996).

Paper IV delves deeper into what makes iconic words so attractive to human brains, with an fNIRS study of Japanese ideophones. We compare brain activity during the processing of Japanese ideophones with activity for verbs and adjectives of matched sensory strength. Despite being rated by participants in behavioural studies as equivalent in sensory strength, ideophones engaged a much more distributed network of cortical regions with relevance to multiple sensory modalities, as well as showing greater activity in areas of multisensory convergence. That the semantic frames of ideophones are rich and multisensory in nature is generally well known (Akita, 2012; Iida & Akita, 2023; Nuckolls, 2019). However, these weren't the only differences between ideophones and other word classes. The processing of ideophones—once described as belonging to the "lunatic fringe of language" (Frankis & van Ostade, 1991)—also elicited increased activation in frontal regions involved in higher-order cognitive control, metacognitive monitoring, and discourse-level planning. This warns against perspectives on iconicity that equate ease (e.g., of learning and processing) with simplicity, and in so doing underestimate the full extent of iconicity's functions in language.

In short, there is more to ideophones than meets the ears, and the goal of the final paper (V) is to explore how documentary linguistics can do justice to this complex word class—especially important given its history of marginalisation. We present a cohesive framework for documenting, analysing, and showcasing ideophones in a way that works for their unique semiotic properties. Our mixed methods approach combines tried and true strategies from documentary linguistics with new and experimental approaches to the study of meaning from psycholinguistics. The paper concludes by highlighting some of the possibilities digital media offer for rethinking the traditional dictionary structure to better reflect the organisation of meaning in ideophone lexicons.

2. What does it mean for language to be motivated? And what doesn't it mean?

"If all the world were to call you
Hermogenes, that would
not be your name."

Plato, Cratylus, 383a

Cratylus teases Hermogenes with this statement in the opening lines of Plato's dialogue, but the deterministic attitude behind his teasing is accurate of the way motivation in language has historically been viewed, and its legacy lives on in how iconicity is illustrated and often how it is operationalised in modern studies. In this section, I highlight how perspectives on iconicity have evolved in recent times, focusing the discussion around three prominent 'myths' about iconicity which the findings in this thesis call into question. The first of these is the myth of universality; the idea that there is one 'perfect' form for the expression of every meaning (as Cratylus suggests). I show that iconicity is really much more flexible than this, which is just as well, since universalist perspectives on iconicity can be quite limiting. Clarifying the flexibility of iconicity brings us to the second myth, which is that iconicity is limited in the kind of information that it can express. Papers IV and V in particular highlight that the expressivity afforded to ideophones as iconic and systematic words goes far beyond simple perceptuomotor analogies. The third and final myth has to do with the idea of iconicity as exceptional. Several papers in the thesis deal with ideophones, and compare them with words from other lexical classes. I show that the boundaries between prototypical iconic words, like ideophones, and regular lexical items are more fluid than is often assumed, and highlight that iconicity is an everyday part of language of all kinds, albeit on a subtler level. Finally, I consider what this revised view of iconicity means for theories of language structure, particularly regarding the role of arbitrariness. As the first paper in the thesis contains an in-depth explanation of relevant theoretical concepts for this discussion, I do not go over these again here but refer readers to Paper I and to the Glossary which summarises my use of key theoretical terms.

2.1 The myth of universality

Many of the earliest investigations of iconicity in language were motivated by a conviction of the existence of "true names" for every concept, the discovery of which would lead to the recovery of a perfect and universally transparent language such as described in the Tower of Babel and similar myths (Eco, 1997). This has carried over to present-day approaches that search for iconicity by searching for universality (see Johansson, 2026 for an overview), or even define iconicity by its universality (Thompson & Do, 2019). It is implied in the methods used in experimental research on sound symbolism, that look for recurring trends in responses to forced-choice tasks (Sapir, 1929; Köhler, 1929; Ramachandran & Hubbard, 2001; Ćwiek et al., 2022)—also how iconicity is measured in Papers II and III of this thesis. This is not to say that these methods of studying iconicity are problematic in themselves. The problem arises when operationalisations of iconicity are not kept distinct from iconicity as a theoretical construct, as this can lead to a deterministic view of iconicity that is not reflective of how it manifests in natural language. In Paper I, I argue that this deterministic view of iconicity is a misconception, that there's a reason all those searching for the perfect language saw their own language as closest to the ideal, and that it's not only because they were deluded by nationalism, but also because, contrary to how it's often perceived, motivation in language is pluralistic, and this stems from the nature of iconicity itself.

Iconicity is about similarity (Peirce, 1932). To recognise that two things are similar, you need to be able to ignore all the ways that they are different, and for this reason iconicity is also, by necessity, selective (Emmorey, 2014; Winter et al., 2023; Punselie et al., 2024). It is the selective nature of iconicity that also makes it inherently pluralistic, as is most evident when we look at iconicity in natural languages. Diffloth (1994) provides a striking example of the plurality of iconic construals afforded by speech in the domain of size-sound symbolism in Bahnar, a Mon-Khmer language with a rich and extensive class of expressives, also known as ideophones. Ideophones are words that depict rather than simply describe their meanings, and "invite and afford" iconicity (Dingemanse, 2019). Papers IV and V deal with ideophones in detail, with very different methodological approaches, but with the shared goal of showcasing these words' incredible expressivity. Examples of Bahnar ideophones include *blɔɔŋ-blɛɛw*, which depicts the numerous reflections caused by a single ray of light on a small, shiny object; *blɛɛl-blɛɛl*, depicting the vivid, intermittent flashes of a small fire; and *blɛɛl-ɲɛp*, depicting the last, intermittent flashes of a small fire as it dies out.

To a Bahnar speaker, there is no more natural way to express these experiences, yet the meanings of these ideophones are far from transparent. To understand them, as Diffloth says, "one must learn to focus on those specific resemblances between sound and meaning which its speakers have selected and exploited" (Diffloth, 1994, p. 109). This highlights that conventionalisation is not only a hallmark of arbitrariness, but is available to and exploited by iconic words as well (cf. Little et al., 2017). One of the most interesting features of Bahnar expressives is the way in which they map vowel height to the semantic dimension of size, shown in Table 2.1.1 (reproduced from Diffloth, 1994, p. 112). Diffloth provides numerous examples illustrating the productivity of these mappings. For instance, to depict reflections caused by a larger object than that which scatters light *blɔɔŋ-blɛɛw*, there is the form *blooŋ-blɛɛw*; a small fire flashes *blɛɛl-blɛɛl*, but a larger fire flashes *bleel-bleel*, etc. The highest vowels are used for very large things, as with *cəwiir* which depicts a very large, twisted mouth (cf. *cəwɛɛr* for a small mouth) (Diffloth, 1994, p. 112).

Table 2.1.1 Vowel-size mappings in Bahnar expressives (reproduced from Diffloth, 1994, p.112)

ENORMOUS	ii	uu	i	u
BIG	ee	oo	e	o
SMALL	ɛɛ	ɔɔ	ɛ	ɔ

The important thing to note about these examples is of course that they directly contradict the large body of comparative and experimental research on size sound symbolism, which shows that people usually prefer to map high vowels to smallness, and low vowels to largeness (Jespersen, 1922b; Sapir, 1929; Ultan, 1978; Shinohara & Kawahara, 2010; Ohala, 1994; Ekström, 2022). Diffloth speculated that Bahnar speakers may attend more to the way in which the tongue occupies a larger volume in the mouth for high vowels compared to low ones. Later, Tsur (2006) argued that they may attend more to the frequencies of the first rather than the second formant, perhaps via manipulations of voice quality, which are common in expressives (see Akita, 2021b for some relevant discussion). Regardless of how you analyse it, Bahnar ideophones demonstrate the flexibility with which people can entertain different iconic construals for the same sound.

Universalist perspectives on iconicity are so prevalent that typologies of phonosemantic associations are almost non-existent. However, notable exceptions include a recent typology of sound symbolism in proximal and distal demonstratives, which revealed as many as 12 different sound symbolic strategies for representing just this single two-way contrast (Dahlgren & Kittilä, 2026). Research on iconicity has for so long been focused on proving its existence through universal trends, that the question of how to map diversity in the realisation of iconicity is only now beginning to be asked. However, it's clear that there are quite a few steps in between total arbitrariness and the single, mythical iconic ideal, with iconicity affording considerably more flexibility than is generally credited to it.

The flexibility of iconicity is a recurring theme across many of the papers in the thesis. Paper I frames construals of iconicity as fluid, and demonstrates this by illustrating their susceptibility to shift under the influence of different contextual factors. In Paper II, we find that English speakers match certain Japanese ideophones to their meanings at rates considerably *below* as well as at or above chance, hinting at conflicting iconic associations. While in Paper III, we see different dyads converging on different iconic forms for the same meanings across the eight independent transmission chains, perhaps depending on the current alignment of forms in the rest of the language. Strikingly, less than half of the demonstrably iconic mappings in the study recreated the one true 'perfect' mapping for which the vocalisation was originally created.

Research on iconicity is often geared towards demonstrating universality, but iconicity does not need to be universal to exert a structuring influence on language; in an evolutionary context, small biases in relevant populations at the relevant time are enough (Kirby et al., 2014). In fact, it is arguably better for iconicity to be somewhat flexible, since language is an ever-changing target to which it must adapt. Consider how quick proponents of arbitrariness are to point out how a 'perfect' iconic language would soon be undone by the inevitable processes of sound and meaning change constantly operating on linguistic signs (e.g. Monaghan et al., 2011).

2.2 The myth of limitations

The flexibility of iconicity can also be seen as among the sources of its incredible expressive power, as the same form can be used to convey a variety of different effects (Sasamoto & Jackson, 2016; Winter et al., 2019; Malysheva, 2025a; Akita et al., 2024; Kumagai, 2020; Winter et al., 2021; Werner & Kaplan, 1963). Malysheva, in her analysis of English phonaesthemes, describes iconicity as a force that is constantly "expanding through phonaesthemic

groups trigger[ing] further sensory associations" (Malysheva, 2025a, p. 137). For instance, the English phonaestheme *bl-* supports such diverse meanings as *blare*, *blow*, *blizzard*, and *blush*. As argued in Akita et al. (2024), the semantic networks introduced by iconicity are distinct from those of metaphor or metonymy. Winter et al. (2021) suggest that "pluripotentiality" of association is a core feature of sound symbolism (see also Werner and Kaplan, 1963, who talk of "plurisignificance"). Sasamoto & Jackson (2016) demonstrate how Relevance Theory (Sperber & Wilson, 1986) can be used to explain how speakers resolve the relevant iconic associations from among many potential ones in context, in a study of Japanese onomatopoeia (see also Tsujimura, 2016 for more examples of the semantic flexibility of Japanese ideophones). In Paper IV, we found regions of the brain involved in conflict monitoring to be particularly active during the processing of Japanese ideophones—more so than for less iconic words of matched sensory profiles. Again, this reflects that iconic words support a wider diversity of meanings.

Despite this, iconicity is often seen as inherently limited when compared with arbitrariness. For instance, Hockett (1960), in introducing his influential "design features" of language, had this to say: "The design-feature of "arbitrariness" has the disadvantage of being arbitrary, but the great advantage that there is no limit to what can be communicated about." Dingemanse et al. (2015, p. 611) are more explicit in specifying a role for arbitrariness in enabling communication about "concepts for which direct perceptual grounding is unlikely to be available."

One theory as to why language is not more iconic is that iconic words are too strongly linked to specific referents, so are unlikely to undergo abstraction (Lupyan & Winter, 2018). For instance, an onomatopoeia which imitates a dog's bark may not work well as a general term for dog since it puts people too much in mind of a particular type of dog—the kind of dog whose barks match the onomatopoeia—making it hard to generalise to other types of dogs with different barks (Edmiston & Lupyan, 2015). Still, as this thesis demonstrates, there is a considerable diversity of ways in which spoken language can be iconic beyond direct sound imitation. In studies involving a less direct form of iconicity, sound-shape symbolism, iconic labels actually facilitated categorisation relative to anti-iconic and arbitrary ones—presumably by exaggerating category-diagnostic features (Lupyan & Casasanto, 2015; Scott et al., 2025).

The link between iconicity and referential specificity may vary depending on the degree or type of iconicity. For instance, an investigation of iconic signs in ASL found that while weakly iconic signs had few semantic neighbours, highly iconic signs had many. Similarly, Japanese ideophones have alternately

been described as having highly specific meanings restricted to limited contexts (e.g. Akita, 2012), but also highly flexible meanings in diverse contexts (Tsujimura, 2016; see also Nuckolls, 2026 for similar examples in Pastaza Kichwa). In fact, descriptions of ideophone semantics range from "[having] the semantic content of entire clauses" (Beck, 2008, p. 38) to "semantically empty and context dependent" (Moshi, 1993, p. 190). While the latter statement has been criticised (Dewey et al., 2024), it's argued that while the meanings of some ideophones are indeed highly specific, others are "underspecified by design" (Nuckolls, 2026, p. 474; Tsujimura, 2016). The point being, not all iconic words are imitations of specific things in the world. They can also imitate general abstracted properties, which taken together would allow iconicity to serve both the functions of individuation (e.g. Lu & Goldin-Meadow, 2018) and categorisation (e.g. Scott et al., 2025). We see evidence of both types of iconicity in the vocalisations in Paper III. Some are clearly imitations of specific referents (e.g., *tiger*), while others simply pick out a general quality and exaggerate it (the vocalisations *sharp*, *big*, and *dull* are good examples). The latter are much more versatile, showing up as attractors for a variety of meanings. They are particularly common in vocalisations for property concepts, suggesting speakers adapt their choice of iconic strategy according to the type of information to be conveyed (see Schiefner, 2025 for similar observations about iconic strategies used for abstract and concrete concepts in sign languages).

Going further, although most research on sound symbolism (in keeping with research on cognition generally) has been conducted in concrete domains, there are reasons to think iconicity may not be as incompatible with abstract meanings as commonly assumed. First, one account of the flexibility of some iconic associations is that they tap into higher-order semantic factors relevant to a diversity of concepts. One study had participants rate non-words and images on a variety of semantic scales, and found that their responses clustered into four higher-order factors which they termed activity, valence, potency, and novelty (Sidhu, Vigliocco, et al., 2022). They also showed in a forced-choice task that participants were more likely to match nonwords and images associated with the same higher-order factors, and speculated that these higher-order factors may be involved with iconic associations of more abstract types (see also Aryani et al., 2020). Some abstract domains for which sound symbolic affects have been demonstrated include the domain of *health* (Motoki et al., 2021), *personality* (Shinohara & Kawahara, 2013; Sidhu et al., 2019; Uno et al., 2020), *mood* (Yu et al., 2021), *luxury* (Motoki et al., 2023), and *efficacy* (Park et al., 2021).

Second, iconicity goes hand in hand with systematicity, providing a pathway toward abstraction. By systematicity, I refer to statistical relations between patterns of word forms and meanings in lexicons (see Paper I; Dingemanse et al., 2015). Systematicity is another way in which form-meaning pairings in lexicons can be motivated. It is distinct from iconicity in that it does not imply a direct resemblance between form and meaning. Rather, the association between meaning and form depends on similar forms being used for similar meanings elsewhere in the lexicon. The versatility of iconicity is such that iconic mappings tend to also become systematic in lexicons as they are reused across different concepts, with phonaesthemes being a prime example (Bolinger, 1950; Carling & Johansson, 2014). As more and more concepts are drawn into the orbit of a particular sound pattern, this could bring any higher-order factors connecting them into relief, facilitating the transfer of the pattern to more abstract domains.

Third, while perceptuomotor experiences are central to the meanings of many iconic words, they are not their only source of meaning. This is particularly clear from the exploration of ideophones in papers IV and V. The neurological data in Paper IV shows that the processing of Japanese ideophones does not only involve perceptual brain regions, but also regions relevant to social and emotional processing. Paper V, which tackles the challenges ideophones pose to lexicographers, emphasises how the meanings of ideophones often are not self-contained but emerge dynamically from how they systematically relate to the meanings of other ideophones in their wider semantic and phonological networks. Both papers IV and V emphasise the breadth and complexity of ideophone meanings, which frequently evoke complete scenes or scenarios involving multiple elements. These features of ideophone meanings—their social and emotional elements, internal grounding in language, and situational content—are more often linked to abstract than concrete concepts (Borghi et al., 2018; Desai et al., 2018), and indeed many ideophones do convey meaning of an abstract nature, in addition to their more concrete sensory meanings. For example, ideophones in Mundari are used to convey moral judgment and index social types (Badenoch et al., 2019; Choksi, 2020; see also Alexandre, 1966 for similar observations on Bulu), while ideophones in Pastaza Kichwa are used to reason about the true nature of situations or portend future events (Nuckolls, 2019). Many now also acknowledge that the distinction between concrete and abstract concepts is far from clear-cut, and that sources of knowledge about both types of concept overlap (Marques & Nunes, 2012; Barsalou et al., 2018; Löhr, 2024; Diveica et al., 2025; Banks et al., 2023). Rather than placing concepts on a spectrum from concrete to abstract, researchers increasingly see them as occupying different points on a shared

multidimensional space, under which view iconic words' association with one type of knowledge (i.e., concrete perceptual knowledge) should not preclude them from also encoding information of other kinds, as the examples with ideophones here demonstrate.

Finally, as argued in Hodge and Ferrara (2022), iconicity is not limited to material perceptual resemblances, but also includes resemblances of relation (as in relative and diagrammatic iconicity) and association (as in metaphoric iconicity), with the latter two types of iconicity being considerably more abstract. For example, the iconicity of English *glib*, related to the archaic *glibbery* "slippery", in constructions like "Politicians need to do more than provide *glib* answers to difficult questions" is of an associative type. Colour terms in Korean make use of relative iconicity (Rhee, 2019), while diagrammatic iconicity is often found in grammatical constructions (Haiman, 1980), conveying abstract relational meaning through the way in which affixes bind to verb stems (Bybee, 1985), or via asymmetries in possessive constructions (Devylder, 2018).

When systematicity is added to the mix, motivated patterns may come to pervade even more abstract corners of the lexicon. Examples from English include kinship terms (where irregular sound changes in *father* and *mother* have brought them into alignment with *brother*), demonstratives and other words encoding relations (the only words in English to occur with an initial voiced ð, e.g., *the*, *this*, *that*, *then*, *there*, *thus*, *though*, etc.), and question words (occurring with initial *w-* or in some dialects the more marked *hw-*) (examples taken from Jakobson, 1965; Jakobson & Waugh, 1979; Waugh, 1994). Among the words picked out as phonaesthemic by a model trained to identify form-meaning regularities from English word embeddings were such abstract examples as *truism*, *celibate*, and *oblique* (Liu et al., 2018). Paper I provides a full typology of the many ways in which form-meaning relations in lexicons can be motivated, including various forms of iconicity along with indexicality and systematicity. When typical examples of iconicity are considered as part of this broader wheelhouse of motivation, we get a sense of just how extensive patterns of motivation in lexicons really are (Sidhu, 2025).

2.3 The myth of exceptionalism

The final point I wish to highlight in relation to the findings presented in this thesis is that, similar to the concrete/abstract (in)distinction, there is no clear boundary between iconic and non-iconic vocabulary in lexicons. In Paper II, which compares guessing accuracies and iconicity ratings for ideophones and other items of vocabulary, a majority of words from *both* groups were

guessed at rates higher than chance—a finding we take as indicative of their iconicity. When presenting the paper, audiences often found this result surprising. They expected the ideophones to be iconic, but not the regular lexical items. The iconicity ratings revealed participants had the same expectations; they consistently underestimated the iconicity of non-ideophones, rating them lower in iconicity than ideophones guessed at the same accuracy levels. Figure 2.3.1 provides a neat visual summary of this effect, in a meme from 2021 in which a flamboyant Taron Egerton as Elton John carpools with the more muted Richard Madden. This version of the meme is taken from a tweet by Mark Dingemanse, who uses it to illustrate how ideophones, which are structurally marked, overshadow less ostensive, 'everyday' forms of iconicity, like phonaesthemes.



Figure 2.3.1 "Ideophones were made for this." Meme from 2021 of the flamboyant Taron Egerton as Elton John carpooling with the more muted Richard Madden, taken from a tweet by Mark Dingemanse who uses the meme to illustrate how ideophones overshadow other iconic phenomena like phonaesthemes.

The iconicity of words like ideophones attracts so much attention that it's natural and convenient to use them as examples in discussions of iconicity. The downside to this is it can create an impression of iconicity as being separate from so-called 'regular' language, simply because words like ideophones tend to separate themselves and to operate differently from other parts of speech (see Papers IV and V). We start to think of iconicity as just a feature of marked, 'funny' words (Dingemanse & Thompson, 2020), and so the iconicity of everyday, 'unfunny' language flies under the radar. What really separates ideophones from other words, however, is *not* that they are iconic, but rather that they are depictions (Dingemanse, 2019).

The distinction between iconicity and depiction is subtle and often overlooked, but one way to think of it is that iconicity is a way of seeing something, whereas depiction is a way of presenting something (Clark & Gerrig, 1990; Clark, 1996). Depiction is what helps us to "see" a particular form as iconic. Depictions are deliberately performative; they draw attention to themselves to signal that their interpretation requires some imagination on the part of the audience, to see the depiction as more than what it ostensibly is (Dingemanse, 2015; Akita, 2021a). An example can be found in the song *Tata Teutonne* ('German aunt') by Serge Gainsbourg from his album *Rock Around the Bunker*, a satirical, rockabilly rendition of his experiences as a Jewish child in Nazi occupied France. The lyrics of *Tata teutonne* are full of alliteration *("Otto est une tata teutonne, pleine de tics et de totos"*), and Gainsbourg articulates them in a rapid staccato, his marked articulation signalling to the audience that they should be interpreted not just as (descriptive) words but also as a depiction of machine gun fire. In fact, their descriptive content is arguably secondary to their depictive effects in this song, with Gainsbourg favouring obscure and archaic terms that maintain the alliteration over less semantically opaque alternatives that would disrupt it. The song's chorus is also a more ostensive depiction of gunfire ("*ratatatatatata ratatatata")*.

Face-to-face interactions are full of depictions, which make use of resources like gesture, prosody, onomatopoeia and ideophones, but also, like *Tata Teutonne*, ordinary language (Ferrara & Hodge, 2018). For example, imagine someone is cutting you a piece of cake, and you only want a small slice. You might express this by asking for "just the *teeeeeeniest sliver*" while making a pinching gesture; lengthening and accentuating the high vowel in *teeny*; using a markedly higher pitch for "*teeeeeeniest*"; and accentuating the first syllable of "*sliver*". In this way, depictive constructions highlight the iconicity that is already present in everyday vocabulary, but which can be easy to overlook outside the context of situated language use.

Ideophones are special because, unlike other words, depiction is their *modus operandi*. However, they too often come from ordinary language, sharing derivational relationships with regular lexical stock in other word classes (see examples in Childs, 1989; Hamano, 1998, p. 52; Dingemanse, 2019, 2021). Paper IV also shows that, for all their special properties, there is no categorical difference in the processing of ideophones versus words of other classes. Previous studies comparing ideophones and non-ideophones reported bilateral patterns of activation reflective of enhanced embodied processing only for ideophones (Arata et al., 2010; Hashimoto et al., 2006), but Paper IV shows that when controlling for perceptual strength, bilateral activation patterns are

also found for regular vocabulary (albeit less extensively). The point here is that iconicity itself isn't special; it is only special to recognise it as such.

2.4 Implications for the structure of language

Despite the prevalence of iconicity and other forms of motivation in lexicons, there are also many words where, at least synchronically, there is no clear link between meaning and form. This absence of motivation has typically been afforded an equal significance to its presence in theories of language structure (e.g. Monaghan et al., 2011, 2014; Perniss et al., 2010; Perniss & Vigliocco, 2014; Dingemanse et al., 2015). Both have been framed as features rather than bugs. In the case of motivation, we have clear evidence that it is a feature. The empirical data attests that iconicity has observable benefits for learning, processing, and communication (Perniss et al., 2010; Perniss & Vigliocco, 2014; Dingemanse et al., 2015; Nielsen & Dingemanse, 2021; Murgiano et al., 2021; Barker & Bozic, 2024; Oko & Mbey, 2025). Arbitrariness, on the other hand, is often claimed to be key to language's expressive power, yet few provide empirical evidence to back this up. Those who have attempted to demonstrate the advantages of arbitrariness in the structure of vocabulary tend to do so via comparisons to a model of motivation that is unrealistic.

For example, one of the most commonly proposed advantages of arbitrariness is that it makes semantically related words discriminable, particularly in large lexicons. The empirical evidence for this comes from two studies comparing the ability of computational models to learn artificial vocabularies where the relations between word forms and meanings were either arbitrary or motivated in some way (Gasser, 2004; Monaghan et al., 2011). One of the studies (Monaghan et al., 2011) also tested the vocabularies on university students. However, in both cases the model used for the motivated vocabulary was a poor reflection of what motivation in natural language really looks like.

In the first study (Gasser, 2004), connectionist networks learned form-meaning pairings varying along three dimensions. For the arbitrary language, the values for each dimension were randomly generated, whereas in the motivated language they were highly correlated (the values for the form and meaning were identical along two out of the three dimensions). With small lexicons the networks did better in the correlated condition, but with larger lexicons they did better in the random condition, presumably as there was less overlap in the form-meaning pairings, keeping words discriminable. However, the model failed to capture two key properties of iconicity: its pluralism and its selectiveness (Section 2.1). As we have seen, there is often more than one way in which form and meaning can be seen

to correspond in iconic words, and iconicity also tends to be highly selective, singling out just a few specific and salient formal and semantic features to correlate among many possible ones (Emmorey, 2014; Winter et al., 2023; Punselie et al., 2024). In Paper III, we see an example of how this property of selective salience can make iconic mappings learnable even in cases where the forms overlap (with *eat*, *meat*, and *berries*). Figure 2.4.1 illustrates how under a pluralistic view of iconicity as involving the selective construal of specific form-meaning relations from among many possible ones, the problem Gasser models disappears.

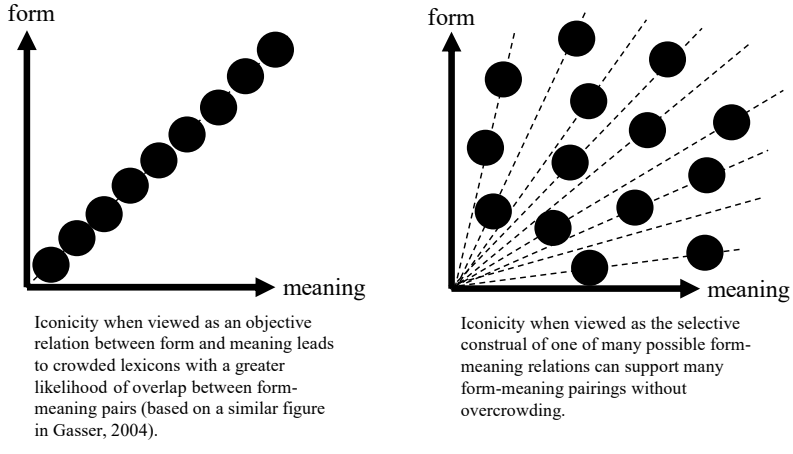


Figure 2.4.1. Comparison of deterministic (left) versus pluralistic (right) perspectives on iconicity. The left panel is based on a similar figure in Gasser (2004), while the right panel shows how a pluralistic model of iconicity is more scalable

The second study, by Monaghan and colleagues (2011), also argues that arbitrariness is preferable for discriminating meanings, but again the comparison condition does not do justice to the alternatives to arbitrariness offered by natural languages. The artificial lexicon in this study had two categories of words, and the model was tested on its ability to learn both the individual words and their category. All the words were of a CVC structure, with half made up of fricatives and front vowels (e.g. *fiʒ*, *ʒif*), and the other half plosives and back vowels (e.g. *ka:k*, *gu:k*). In the arbitrary condition, words were assigned to categories such that half the words in the category were of the fricative + front vowel type, while the other half were of the plosive + back vowel type. In the comparison condition, called the systematic condition, all words in the same category were of the same type (either all fricative + front vowel, or all plosive

+ back vowel). Within each category for both conditions, the assignment of individual words to meanings was random. The results showed that learning individual word-meaning mappings was easier in the arbitrary condition, although only when context was provided that disambiguated which category the word belonged to (i.e., the same information provided by the form of the words in the systematic condition). Category learning was always better in the systematic condition. However, the best performance was obtained when only part of the word systematically encoded category information and the rest of the word was arbitrary. The title of the paper highlighted the 'learning advantages' of arbitrariness, without mentioning systematicity.

An issue with the study is that it does not account for the full range of possibilities offered by a motivated account of language structure. When we consider signs produced by human participants under similar demands, these often combine systematicity with iconicity rather than arbitrariness (Verhoef et al., 2015; Motamedi et al., 2019; Winters et al., 2015), as do human languages (Sidhu, 2025). For instance, Winters et al. (2015) provide an example of an artificial language evolved by participant dyads performing a communicative task that required distinguishing both within and between meaning categories. The authors emphasise how this condition uniquely resulted in the development of systematic languages, but the example language they provide is in many ways iconic as well. The meanings in the language corresponded to eight alien figures. Half were blob-shaped, the other half were star-shaped, and each also had one distinguishing idiosyncratic feature. The evolved terms for the blob aliens in the example provided of a language developed in the mixed condition (with both categorisation and individuation requirements) were *muwu*, *muwumu*, *muwumuwu*, and *meheme*; while the stars were *hapa*, *hupa*, *hopa* and *hepa*. There is an obvious systematicity here, which also taps into iconic distinctions between sonorants and obstruents, and rounded and unrounded vowels, when applied to the domain of sound-shape symbolism (McCormick et al., 2015). Iconicity also appears to have been recruited to distinguish between different aliens of the same type, particularly in the words for the blob aliens, many of which exhibit diagrammatic iconicity. The unmarked form seems to be *muwu*. Partial reduplication (*muwumu*) is used for the alien that has a (single) tail, while full reduplication (*muwumuwu*) is used for the alien that has many legs. *Meheme* is used for the least blob-like of the blob aliens, which has two skinny arms stretched out on either side of its body. The shift to an unrounded vowel, as well as the symmetrical structure of this form can again be construed as iconic. The star aliens are distinguished by markings on their points: one has triangles on its points (*hapa*), one has blobby shapes (*hupa*), one has semi-circles

(*hopa*), and one has lots of little dots (*hepa*). The names for the star aliens are simple and minimally distinctive, and the case for iconicity is less strong here. Still, there are patterns in the assignment of forms to meanings that don't seem accidental, like the unmarked form (*hapa*) being used for the star annotated with triangles on its triangles, or the star with the blobby markings (*hupa*) being distinguished by the same vowel used for blobby aliens—solutions which seem preferable to 'arbitrary' alternatives.

In more natural language contexts, people often make use of the flexibility of iconicity to discriminate within category boundaries. For example, when needing to distinguish among objects belonging to the same semantic category, signers iconically embellish lexical signs to depict imagistic features of the object that aid in its identification (Lu & Goldin-Meadow, 2018). Korean has just five basic colour categories, but well over 500 colour terms that can be used to make further distinctions within these categories *because of*—not despite—its use of iconicity and systematicity (Rhee, 2019). Still, sentiments like the following, from an influential review paper on the role of iconicity in language, are oft-repeated:

> Specifically, iconicity would be favoured by those processes engaged in ensuring that communication is meaningful, in the sense of related to and grounded in our experience; arbitrariness would, instead, be favoured by those processes engaged in ensuring that the linguistic signal is efficient and discriminable, contributing to exemplar learning and the ability to carry out within-category discrimination
>
> (Perniss & Vigliocco, 2014, p. 2)

The authors actually cite Monaghan et al.'s (2011) study here, conflating iconicity and systematicity. What's worse is that in referencing this study, in particular, they conflate it with a concept of systematicity that doesn't even exist in natural languages. The introduction to Monaghan et al.'s study identifies their model of systematicity as based on John Wilkin's (1668) attempt at creating a "perfect" universal language. Wilkins's approach was similar to that which would later be used in the scientific naming of plants and animals. He divided the world into broad concepts, assigned a syllable to each concept (randomly), then made further divisions within each concept using the following consonant and vowel. For example, *de* meant element; *deb*, the first element, fire; *deba* a flame; *debi*, a spark; *debo*, lightning, etc. It's difficult to exaggerate just how far-removed this deterministic, top-down approach to systematicity is from how systematicity manifests in real languages via phenomena like phonaesthemes, that grow organically out of existing patterns in

words, are indeterminate by nature, and, as we've seen, work in tandem with iconicity (Carling & Johansson, 2014; Malysheva, 2025a; Sadowski, 2001).

The idea that iconicity is less 'efficient' than arbitrariness also deserves some critical reflection. The perspective stems from observations on the historical development of sign languages (Frishberg, 1975), and on the evolution of graphical symbol systems (Garrod et al., 2007), where it's been noted that as signs evolve to become more efficient (from a production/perception standpoint), they often do away with iconic elements. However, efficiency should be considered as relative to the amount of information conveyed; it's only increased if no information is lost in the process of reducing iconic elements. While a reduction in extraneous or elaborated iconic elements likely does contribute to efficiency early on in the development of signs, it's also widely documented that signers frequently *(re-)embellish* signs with additional iconic elements in language use, where these iconic embellishments serve to communicate additional information (e.g. Ferrara & Halvorsen, 2017; Ferrara & Hodge, 2018; Lu & Goldin-Meadow, 2018).

Blanket statements about the efficiency of arbitrariness relative to iconicity also seem quite unfair when considered in the context of research on the semantics of ideophones. The semantic density of ideophone meanings is a key theme in both papers IV and V. Paper IV provides empirical evidence that ideophones, thanks to their iconicity, convey much richer perceptual, social, and affective information than other sensory words of comparable perceptual strength. Paper V tackles the challenges the semantic richness of ideophones poses for documentary linguists. Consider the following observation from Diffloth (1972):

> In trying to paraphrase an ideophone with ordinary words of the same language, we find that several sentences are often needed, and, even then, the paraphrase is not wholly satisfactory.
>
> (Diffloth, 1972, p. 441)

Nicholas Evans, in his introduction to the ground-breaking *Dictionary of Mundari Expressives* (Osada et al., 2019), describes how ideophones "crystallis[e] recognisable, experientially recurrent chunks of reality in ways that are both familiar to an English speaker and at the same time difficult to express with concision." Consider the following definition for *epaṛa-ipiṛi*, below:

> (1) the habit in two persons of omitting some work with the intention that it should or will be done with the other; or of putting off, now the one, then the other, a work which should be by the two together, so that

> the work remains undone. (2) the fact of two yoked bullocks being caused to advance, now the one, then the other, alternately, so that the cart or whatever they have to draw, does not start.
>
> (Osada et al., 2019, p. 106)

Clearly, the solutions offered by an arbitrary vocabulary for the description of some experiences are painfully inefficient when compared to those a motivated, depictive vocabulary affords.

Other claims about the advantages of arbitrariness prove similarly weak when set against a proper consideration of motivation. For instance, consider the following (from another review paper):

> Studies of the cognitive functions of language have shown that arbitrary labels facilitate learning of type/token distinctions (e.g., the general concept 'dog' versus a specific instance of a dog such as Fido) [108], and, in comparison to iconic expressions, are more effective at activating such conceptual states [109,110], possibly because iconic forms necessarily activate more specific instances while arbitrary forms activate a more general and abstract representation [111].
>
> (Dingemanse et al., 2015, p. 611)

The studies cited in support of these claims, however, compare linguistic labels with non-linguistic cues (Lupyan, 2012; Lupyan & Thompson-Schill, 2012; Boutonnet & Lupyan, 2015; Edmiston & Lupyan, 2015). For example, the noun 'dog' versus the sound of a dog barking, or the novel animal name 'shonk' versus a novel sound produced by animals called 'shonks' (Lupyan & Thompson-Schill, 2012). They do not compare arbitrary linguistic labels with motivated ones[1]. As is clear from Paper IV, iconic words are processed using the same linguistic regions of the brain as other words; we do not treat them as non-linguistic (see also Hashimoto et al., 2006). Moreover, studies since have shown iconic labels can actually lead to *better* categorisation than arbitrary ones by highlighting category-relevant features (Scott et al., 2025; Li et al., 2026). Other evidence from research on sign languages suggests that enhanced conceptual activation for iconic words is task-specific,

[1] One of the experiments in Lupyan and Thompsan-Shill (2012) compares nouns (e.g., "dog") with verbs that happen to be onomatopoeic (e.g., "barking"), but the purpose of this was to compare words that refer to categories (i.e., linguistic *labels* specifically) with words that refer to properties, to show that the effect is not simply due to the word status of the nouns but that it relates to their function as linguistic labels. They do not compare iconic versus arbitrary labels specifically.

occurring only when the eliciting stimulus exhibits relevant features (McGarry et al., 2023; Ortega & Ostarek, 2021). Importantly, however, there is no evidence to suggest that iconic signs are any less effective, or non-iconic signs any better, at activating conceptual representations. That is, iconic signs' resemblance to things in the world does not hinder their ability to perform the jobs of linguistic labels (in some cases, it even augments it), nor does the unmotivated status of non-iconic signs augment their ability to do the same (c.f. Edmiston & Lupyan, 2015).

Still, there may be some unfairness here as the studies cited above all investigate the labelling of concrete concepts. Arbitrariness, relative to iconicity, has specifically been argued to support meanings that are abstract in the sense that they reside more in the mental realm than in the physical, perceptual one (Lupyan & Winter, 2018). Lupyan & Winter (2018) argue that abstract concepts in this sense may favour arbitrariness, although their evidence is (as they admit) correlational rather than causal and direct. They point to the fact that English words rated lower in iconicity are also rated lower in concreteness, although this is complicated by the way iconicity is defined in these tasks potentially biasing concepts that are concrete (further studies also show the relation does not hold across languages and datasets; Perlman et al., 2018). To address this, they also show that words independently rated as high in iconicity, but low in concreteness, were more likely to be (mis)classified as concrete rather than abstract in a forced-decision task. However, the effect size was small, and the particular statistical analysis used potentially anti-conservative, leading the authors to stress that the finding is "in need of further confirmation" (Lupyan & Winter, 2018, p. 7). The weak and inconsistent results here may partly relate to fundamental issues with the concrete/abstract dichotomy itself (Connell & Lynott, 2012; Barsalou et al., 2018; Löhr, 2024). Given the problems inherent to the concrete/abstract division, it's also not clear what these misclassifications really show about participants' understanding of the words or their ability to access their meanings.

From the other side, there has been interest in the field of foreign language learning as to whether iconic gestures could be used to support the learning of (non-iconic) abstract words. Again, it's not quite the comparison we need as it was not the words themselves that were iconic but only the accompanying gestures. However, if iconicity really were inimical to abstraction as arbitrariness accounts claim, then we might expect iconic gestures to interfere with learning words for abstract concepts. Instead, several studies find that they facilitate learning, even for concepts rated low in concreteness (Macedonia & Knösche, 2011; Repetto et al., 2017; Andrä et al., 2020; Rodríguez-Cuadrado et al., 2023), while one study found no effect (neither facilitatory nor

inhibitory) (Mayer et al., 2017). As discussed in Section 2.2, there is also evidence from research on ideophones (Paper IV, V) that iconic words may not be as incompatible with abstract meanings as generally assumed, though unfortunately this does not have any bearing on whether arbitrary words are comparatively more suited to conveying abstract meanings.

The goal of this discussion is not to provide a definitive answer on the relative importance of motivation and arbitrariness to the structure of vocabulary, but rather to highlight that there are weaknesses in the case that has been made for arbitrariness which come to light when motivation as a real, natural, and powerful phenomenon in language structure is properly considered. The principle of arbitrariness has had such a chokehold on ways of thinking about language structure for so long, virtually every piece of writing on the topic assigns a role to it. Yet the empirical evidence for arbitrariness *specifically* serving some functional role in language is barely there and, on closer inspection, often questionable. Unfortunately, because alternative hypotheses are so little considered, it's just as hard to rule out a role for arbitrariness as it is to claim one. The proper comparisons have not been made. However, it's worth considering the possibility that the arbitrariness of the linguistic sign may itself be arbitrary. As Saussure observed, "no one disputes the principle of the arbitrary nature of the linguistic sign, but it is often easier to discover a truth than to assign it to its proper place" (Saussure, 1916, p. 68). When can arbitrariness in language really be considered a matter of design, preferable to a more motivated solution? This is the question that needs to be answered for arbitrariness, as we are answering it for motivation, so that both facets of language structure can be assigned their proper place.

3. Carving a space for motivation in theories of language change

"Modern linguistic science… is so preoccupied with etymology, with the origin of words, that it pays much more attention to what words have come from than to what they have come to be"

Otto Jespersen, 1922

We've established that iconicity and motivation are a regular part of language structure. The next step is to uncover the processes that have caused this. It's well-established that iconicity is often involved in the formation of words (Carling, 2017; Flaksman, 2021; Carling & Johansson, 2014; Körtvélyessy & Štekauer, 2020; Liberman, 2026; Dressler & Kilani-Schoch, 2026), and that this constitutes a core source of iconicity in lexicons (Flaksman, 2017). Evolutionary work on iconicity has also tended to focus on its role in the emergence and formation of signs (see Dona & Schouwstra, 2026 for a review). However, there is anecdotal evidence that iconicity is involved in the development of words too, both on the meaning and on the form side. This strengthens the argument that iconicity is an important design principle of language, since, as expressed by Fischer (1997, p. 63), "it is via change that the structure of language and the principles underlying it often become more visible."

Fischer (1997) provides an astute analysis of the role of iconicity in syntactic change. Part of the fundamental nature of language as a social institution is that it is constantly changing, but these changes are not always adaptive. As Joe Greenberg once famously put it, "a speaker is like a lousy auto mechanic: every time he fixes something in the language, he screws up something else" (quoted in Croft, 2002, p. 5). Fischer suggests that, as ambiguities emerge due to changes taking place elsewhere in the system, iconic strategies offer language users an attractive pathway back to understanding, thus shaping the way in which the language develops next. She provides two examples of syntactic changes in English which have led to reductions in ambiguity (the spread of

periphrastic ‘do’, and the bleaching of possessive ‘have’ to semi-modal ‘have’), where iconicity offers a more powerful explanation for where and how the change occurred than traditional explanations do.

I focus the discussion here on the role of iconicity in changes operating on the more basic level of the lexicon itself. While the idea that the form of a construction can influence its tendency to develop along a particular path is not so controversial in the realm of grammar and syntax, and iconic accounts of language structure have a strong history here (Givón, 1985; Haiman, 1980, 1985), it is less common to see iconicity incorporated into theories of phonological and semantic change. Indeed, linguists generally assume that word forms and meanings evolve independently, with developments on one side having no bearing on developments on the other. The idea that the meaning of a word could influence its form in particular is problematic for the Neogrammarian model of sound change as a purely phonetic, exceptionless process, though it is compatible with alternative accounts that model sound change as a process of lexical diffusion (Chen & Wang, 1975).

Yet, we do find anecdotal evidence suggesting bidirectional influences between word forms and meanings in language histories. This evidence can be categorised into three different types:

1. Word forms resisting a regular sound change to preserve sound symbolic patterns.
2. Word forms undergoing irregular sound changes to introduce or strengthen sound symbolic patterns.
3. Word meanings shifting under the influence of word forms.

I use sound symbolism here as a cover term for motivated relations of several types (iconic, indexical, systematic; see Paper I). Some examples of changes of each type are provided in Table 3.1. The sporadic appearance of ejectives, and sporadic retention of implosives, in Cholan (Mayan) words where these sounds have an obvious iconic effect (e.g., *pʼəx* ‘to curse’ vs. *ɓes* ‘to stammer’) are nice examples of the first two types of evidence. Malkiel (1990) provides similar examples in Romance languages, while Carling and Johansson (2014) contains examples from the Germanic languages. Flaksman (2013) coined the term “phonosemantic inertia” to describe the retention of iconic sounds in English words like *cuckoo,* which evaded the Great Vowel Shift. Less has been written about iconicity from the perspective of semantic change, but then semantic change is less studied in general (there is less of a framework for studying it), and is already characterised as “highly irregular” (e.g. Hock, 1986, p. 308), so influences of iconicity on semantic change are more

likely to go unnoticed. Paper III goes some way toward addressing this gap by empirically demonstrating the potential for iconicity to motivate semantic shifts, albeit under highly controlled experimental conditions.

Table 3.1 Three types of evidence for the mutual influences between sound and sense in language change, with examples of each type.

	Example	Source
Irregular sound changes introduce sound symbolic patterns	Sporadic appearance of ejective *p'* in Cholan (Mayan) words related to swelling, e.g., *sip'* 'to swell up', *p'us-pat* 'hunchback', *p'ul* 'smallpox', and involving explosive sounds, e.g. *p'əx* 'to curse'. The sound changes here are all irregular.	(Campbell, 1996)
Failure to observe a sound change preserves sound symbolic patterns	Sporadic retention of implosive *ɓ* in Cholan *ɓes* 'to stammer'	
Unusual semantic shifts resulting in sound symbolism	*husky* 'full of husks' > *husky* 'hoarse (in the throat)' *patter* 'to repeat a prayer' > *patter* 'fast, light tapping sound (e.g., of footsteps, rain on the window, etc.) *miniature* 'image painted with vermilion' > *miniature* 'small in size'	(Jespersen, 1922a)

The challenge for incorporating motivation into theories of language change will be moving from observation to prediction, and connecting experimental data of the kind presented in this thesis to naturalistic data from real languages. One of the findings that comes across most strongly in the thesis is that motivation in lexicons is pluralistic (discussed in Section 2.1), which suggests that motivated patterns in lexicons are likely to be shaped by multiple factors. The history of phonaesthemes in related languages may provide a particularly good case study.

For example, the Germanic languages all inherited a set of words originating from the Proto-Germanic base **twai-* 'two'. Malysheva (2025b) describes how the *tw-* cluster from this base has since become very important in the

English sound symbolic system (in words like *twist*, *twirl*, *twiddle*, *twerk*, etc. referring to certain qualities of motion), but is not used sound symbolically in the lexicons of the other Germanic languages. Modern English has well over 1000 words beginning with this cluster, while German has just under 300, and Icelandic and Norwegian less than 100 (Malysheva, 2025b). Malysheva speculates that some factors which may have contributed to these differences include subsequent phonological developments in each of the languages, as well as parallel lexical innovations in English. English maintained the *tw-* cluster, but it became *zw-* in German, and *tv-* in Icelandic and Norwegian. A word that may have been particularly important to the origin of the phonaesthemic group in English is *twine*, corresponding to *tvinna* 'to twine, to twist' in Icelandic, *tvinne* 'to twine, to twist' in Norwegian, and *Zwirn* 'yarn' in German. The base of these words goes back to Proto-Germanic **twai-* 'two', because a twine is made from twisting together two (or more) threads. The formation of *twine* brought the *tw-* cluster into proximity to the meaning 'twisting', a combination which is suggested to have a weak iconicity (based on the results of nonword experiments by Kwon, 2017). However, the iconicity of *tw-* for 'twisting' in English particularly may have been heightened in relation to other patterns in the lexicon. For instance, an analysis of 52 etymologically unrelated English size adjectives revealed [t] to be the consonant most predictive of small size (Winter & Perlman, 2021b). As Malysheva identifies, initial clusters containing [w] are also found in other words related to movement (e.g., *swing*, *sweep*). Both these semantic associations are also compatible with the meaning 'twisting'. The cluster *tw-* was also exploited in other iconic words like *twitter*, which may have helped English speakers to see it as iconic in other contexts as well (see Section 2.3 on depictive framing). In short, it was likely a confluence of fortuitous factors which facilitated the emergence of the *tw-* phonaestheme in English, but not in the other Germanic languages.

A systems approach to cultural evolution is well suited to investigating historical processes which involve multiple interacting factors, as in the example above (Buskell et al., 2019; Jansson et al., 2025). These approaches model the evolution of cultural systems consisting of multiple traits and their interrelations. A key insight of these approaches is that individuals having acquired a set of cultural traits will then *filter* any new traits they acquire according to how they relate to traits they already possess (Jansson et al., 2025). One way to apply this to a model of lexical change that incorporates linguistic motivation is to consider words as bundles of phonetic and semantic traits. When individuals learn a word, this increases the compatibility between the phonetic and semantic traits the word contains (both within and between the two trait types). Knowledge about the world could be modelled as compatibilities

between semantic traits. Some world knowledge could be modelled as innate to all individuals, while other world knowledge might be acquired through exposure to different combinations of traits. Speech production and perception biases could be modelled as innate compatibilities between phonetic traits. Iconicity could be modelled as innate compatibilities between phonetic and semantic traits (as a simplification). These compatibilities act as a filter on the type of words speakers are likely to learn and create, but as they learn words this also alters the compatibilities between existing phonetic and semantic traits while adding new compatibilities, creating new filters. This captures the effects of systematicity in lexicons. The effects of indexicality can be seen in compatibilities between phonetic and semantic traits which emerge through the acquisition of world knowledge (see Paper I). In this way, a speaker's linguistic behaviour comes to reflect compatibilities shaped by the interaction of all these factors in their own personal history.

The model described above is just an example of the kind of model that would capture important aspects of the account of iconicity (or rather, of construals of iconicity) presented in Paper I. It is not intended to be complete and requires further elaboration and refinement. However, some important properties of the model are that individuals' developing semantic knowledge is informed both by the semantics of the world and by semantic relations encoded in language, and that speech production and perception biases begin as innate but are subsequently able to be shaped by experience with language. This should result in a system that, when incorporated into a cultural causal chain (Scott-Phillips et al., 2018; Miton, 2025), undergoes semantic and phonological change in a way that approximates how we think languages change. Making compatibilities between phonetic and semantic traits a part of this system offers a means of exploring the effects of a motivated account of lexical structure on the dynamics of language change.

In Paper III, I suggest that language is disposed to be motivated, even though it is susceptible to arbitrariness. This is something that could be tested by connecting mechanistic accounts of how motivation is thought to structure lexical relations—like the one outlined above—to observed historical trajectories of language change. For example, to test whether a motivated model can explain or predict the growth (or absence) of phenomena like phonaesthemes as emerging from the interaction of general principles of form and meaning change; general cognitive biases (among which we might include a bias to connect form and meaning); phonetic features present in a population of speakers; properties of speakers' physical and social environments; and individual histories.

The challenge is to offer an explanation for both motivated *and* apparently arbitrary or unmotivated patterns in language structure, without needing to posit arbitrariness (i.e., a disconnect between meaning and form) as a so-called 'design feature' in the underlying cognitive architecture of language (cf. Hockett, 1960; Pleyer et al., 2025). Consider the following extract from Sperber (1985), who characterises the differences between dispositions and susceptibilities in an evolutionary framework:

> Some of the effects of our genetic endowment can be described as dispositions, others as susceptibilities, even though the distinction is not always easy to draw. Dispositions have been positively selected in the process of [biological] evolution; susceptibilities are side-effects of dispositions. Susceptibilities which have strong adverse effects on adaptation get eliminated… Susceptibilities which have strong positive effects may, over time, be positively selected and become, therefore, indistinguishable from dispositions. Most susceptibilities, though, have only marginal effects on adaptation; they owe their existence to the selective pressure that has weighed, not on them, but on the disposition of which they are a side-effect. Both dispositions and susceptibilities need appropriate environmental conditions for their ontogenetic development. Dispositions find the appropriate conditions in the environment in which they were phylogenetically developed. Susceptibilities may well reveal themselves only as a result of a change of environmental conditions.
>
> (Sperber, 1985, pp. 80–81)

I believe the distinctions Sperber makes here may offer an apt account of the relative importance and place of motivation (as one of language's dispositions) versus arbitrariness (one of its susceptibilities) in the cognitive architecture of language. Motivated patterns are positively selected for in the process of language evolution. We see this most evidently in Paper III in the enormous positive effects motivated patterns in form-meaning mappings have on their learnability and communicative utility. This would also explain why motivated strategies are preferentially used and beneficial in situations of child language learning (Massaro & Perlman, 2017; Perry et al., 2018; Green & Perlman, 2022; Green et al., 2026a, 2026b; Sidhu, Williamson, et al., 2022; Perry et al., 2021; Gappmayr et al., 2022; Motamedi, Murgiano, et al., 2021; Motamedi et al., 2024; and see Nielsen & Dingemanse, 2021 for a review), as well as in situations where communication is either compromised in some way (e.g. Drijvers & Özyürek, 2017; Krason et al., 2022; Wheeler & Saito, 2022;

Wilms et al., 2022), or where the requirements on communication are higher than usual, e.g., where there are increased requirements for expressivity (Bermúdez, 2020; Dingemanse, 2011; Gnerre, 2018; Klassen, 2000; Nuckolls, 2010; Oyibo, 2025; Pratha et al., 2016; Samarin, 1969; Webster, 2008, 2017) or precision (Akita, 2025; Mihas, 2013; Yasui, 2023).

On the other hand, to the best of my knowledge there is not currently any evidence that there are situations in which an arbitrary language (compared to a motivated one) is easier to learn or facilitates communication and understanding in some way. There is evidence, however, that arbitrariness can be a secondary outcome of changes producing other properties that are of clearer adaptive benefit to learning and communication. For example, Frishberg (1975, p. 700) describes how changes affecting the "symmetry, fluidity, locational displacement, and assimilation" of iconic signs in American Sign Language, "motivated by such familiar principles as ease of articulation and perception", have also resulted in a loss of iconicity or increase in arbitrariness. Experimental investigations of the origins of arbitrariness in novel communication systems identify similar causal factors (Garrod et al., 2007).

Arbitrariness can also be a side-effect of the emergence of (extensive) systematic structure, akin to morphology, which itself results from iconicity becoming more efficient. For example, an experimental study by Theisen et al. (2010, p. 1971) describes how "people generate signs that are non-arbitrary, those signs become arbitrary as they simplify, by chance there are a few signal-meaning regularities, generations of people propagate these regularities… [resulting in] a set of signs which systematically re-uses arbitrary elements." The reference to chance in the above quotation downplays the contributions of iconicity to the emergence of this systematic structure, since the 'chance' of such signal-meaning regularities occurring is made higher by the iconicity of the original language. The systematic reuse of sign elements contributes to arbitrariness in the wider lexicon, as instead of using a (novel) iconic strategy people reuse elements of other signs whose iconicity is no longer apparent in the new context (see further examples in Verhoef et al., 2015; Motamedi et al., 2019; Motamedi, Wolters, et al., 2021). Systematicity in language has clear adaptive benefits for both learning and communication (Kirby et al., 2008; Garrod et al., 2007). This shows how arbitrariness can be an outcome of evolutionary pressures acting on other adaptive features of language (including, ironically, iconicity), without conveying any adaptive benefits in and of itself.

The final of Sperber's observations pertinent to the discussion of motivation and arbitrariness in lexicons is that both dispositions and susceptibilities require appropriate environments for their ontogenetic expression. Sperber

notes that in the case of dispositions, these environments are likely to reflect the conditions of the environment in which they were phylogenetically developed, while susceptibilities may only emerge due to changes in these environmental conditions. It's well-known that the phylogenetic niche for language was multimodal, face-to-face interaction (Levinson & Holler, 2014), which is also where iconicity and other forms of motivation thrive (Murgiano et al., 2021; Ferrara & Hodge, 2018; Hodge & Ferrara, 2022). Early human societies were also organised into much smaller, more tightly-knit social groups, with a high-degree of shared context, both of which have also been shown to facilitate iconicity (Mudd et al., 2022). It's intriguing to consider whether, had language not come so far from its phylogenetic origins—communication now takes place between members of large and diverse social groups, often asynchronously, often in the absence of multimodal cues, and often concerning topics far-removed from the here-and-now—it may not have become so susceptible to arbitrariness as it is today.

4. Conclusions

I have argued, based on the papers in this thesis, that current perspectives on language structure vastly overestimate the role of arbitrariness, while at the same time underestimating the role of motivation. When both concepts are afforded a proper consideration, we find that theoretical accounts of language structure exaggerate the limitations of motivation at the same time as assigning advantages to arbitrariness that are better explained as arising from the interaction of other, more adaptive linguistic properties. That is, there is no need to posit that the structure of language itself must be arbitrary, to account for linguistic arbitrariness. There may well be a need, however, to posit that the structure of language is motivated to account for many of its other properties, including, perhaps ironically, arbitrariness itself. Motivation in language is also, as the papers in the thesis show, clearly adaptive, and should be considered as among language's 'design features.' Arbitrariness, on the other hand, might be better described as an accident.

5. Author Contributions

In this chapter, I list the respective contributions of the individual scholars in the papers with multiple authors: I, II, III, IV, and V.

I McLean and Motamedi discussed the content of the chapter together. McLean wrote the manuscript, with Motamedi providing feedback at each step.

II All three authors discussed the design of the experiments, McLean created the experiments and collected the data. McLean wrote the manuscript, with Dunn and Dingemanse providing feedback at each step.

III All three authors discussed the design of the experiment, McLean created the experiment and collected the data. McLean wrote the manuscript, with Dunn and Dingemanse providing feedback at each step.

IV All three authors discussed the design of the experiment. McLean prepared the stimuli. Akita and Dewey collected the data with participants in Nagoya. Dewey analysed the fNIRS data. McLean and Dewey wrote the introduction and methods sections of the manuscript together, with Dewey describing the fNIRS procedure and McLean describing the selection of stimuli. Dewey wrote the results section. McLean wrote the first draft of the discussion section, with Dewey helping to revise it, and providing further details particularly on the involvement of specific cortical regions. Akita provided feedback at each step.

V McLean and Dingemanse discussed the content of the chapter together. McLean wrote the manuscript, with Dingemanse providing feedback at each step.

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